

CT

CounterTap

Project Plan & Architecture

A two-app digital ordering system for small and medium shops.

QR-based ordering | UPI payments | Real-time order tracking

Platform

Android (Kotlin)

UI

Jetpack Compose

Backend

Firebase

Auth

Google Sign-In

counter-tap-business

Shop Owner App

Manage | Monitor | Earn

counter-tap

Customer App

Scan | Order | Pay



tenants / {tenantId}

name	string	"Raju's Tea Stall"
ownerId	string	Firebase Auth UID
upId	string	"raju@paytm"
phone	string	Contact number
address	string	Shop address
active	boolean	Shop is open

users / {userId}

name	string	From Google profile
email	string	Google account email
role	enum	"owner" "customer"
tenantId	string	Only for owners

products / {productId}

name	string	"Masala Chai"
price	number	In rupees (e.g. 30)
categoryId	string	Ref to category
imageUrl	string	Storage URL
available	boolean	In stock

categories / {categoryId}

name	string	"Beverages", "Snacks"
sortOrder	number	Display position

orders / {orderId}

customerId	string	Firebase Auth UID
customerName	string	From Google profile
items[]	array	{id, name, price, qty}
total	number	Sum in rupees
status	enum	pending->confirmed->ready->done
paymentStatus	enum	unpaid paid
upiTxnId	string	Entered by customer
note	string	Special instructions

counter-tap-business (Owner App)

Auth

● Splash

Check login state, redirect

● Sign In

Google Sign-In -- one tap, free, unlimited

Onboarding

● Shop Setup

Name, address, phone, logo upload

● UPI Setup

Enter UPI ID shown at checkout

Orders Tab

● Live Orders

Real-time list grouped by status, badge count

● Order Detail

Status buttons, customer info, bottom sheet

● Order History

Completed orders, filterable by date

Menu Tab

● Product List

Grouped by category, inline availability toggle

● Add / Edit Product

Name, price, image, category, availability

● Categories

Add, reorder, delete categories

Settings

● Shop Profile

Edit name, address, logo

● UPI Settings

Change UPI ID

counter-tap (Customer App)

Auth

● Splash

Check login state, redirect

● Sign In

Google Sign-In -- name auto-filled

Main Flow

● Home / Scanner

Camera QR scan + recent shops list

● Shop Menu

Products by category, search, Add button

● Cart

Items, qty controls, special note, total

Checkout

● Payment Screen

UPI deep link opens GPay/PhonePe/Paytm

● Confirm Payment

Enter transaction ID, tap I have paid

● Order Tracking

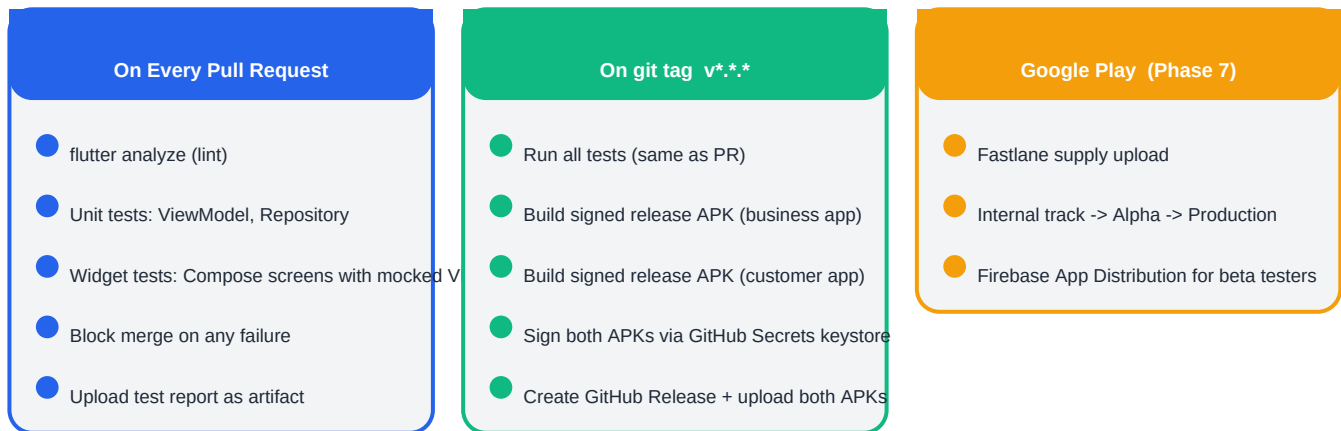
Real-time status stepper, shop phone

History

● Order History

Past orders with re-order button

CI / CD -- GitHub Actions



GitHub Secrets Required

GOOGLE_SERVICES_JSON	Firebase config file (injected at build time)
KEYSTORE_FILE	Base64-encoded Android signing keystore
KEYSTORE_PASSWORD	Keystore password
KEY_ALIAS	Key alias name
KEY_PASSWORD	Key password
PLAY_STORE_JSON_KEY	Google Play service account JSON (Phase 7)

Firebase Cost -- Existing App + CounterTap (10 Shops) | 1 USD = Rs. 95.33

Service	Free Limit	Combined Usage	Monthly Cost
Firestore Reads	50,000 / day	~19,300 / day (39%)	Rs. 0
Firestore Writes	20,000 / day	~6,250 / day (31%)	Rs. 0
Firebase Auth	Unlimited	All users	Rs. 0
FCM Notifications	Unlimited	All orders	Rs. 0
Cloud Storage	5 GB (Blaze)	~60 MB images	Rs. 0
Cloud Functions	2M calls/month (Blaze)	~15,000 / month	Rs. 0
TOTAL			Rs. 0 / month

Upgrade to Blaze before adding images / push notifications -- no monthly fee, pay only for overages.

Phase 1 | Foundation

Week 1-2

- GitHub private monorepo setup
- Both Android projects (Kotlin + Compose)
- Firebase project (dev + prod)
- Hilt DI, Navigation, base architecture
- Google Sign-In in both apps
- Shared data models module
- GitHub Actions PR check workflow

Phase 2 | Business App Core

Week 3-4

- Shop onboarding flow
- Add / edit / delete products and categories
- QR code generation screen
- UPI ID settings

Phase 3 | Customer App Core

Week 5-6

- QR code scanner (ML Kit)
- Menu browsing screen
- Cart functionality
- Order placement

Phase 4 | Real-time Orders

Week 7

- Live order list (Firestore listeners)
- Order status update buttons (owner)
- Order tracking screen (customer)
- FCM push notifications on new order

Phase 5 | UPI Payments

Week 8

- UPI deep link in customer app
- Manual transaction ID confirmation (v1)
- Payment status saved in Firestore order

Phase 6 | CI/CD + Release

Week 9

- Release GitHub Actions workflow
- Keystore setup + APK signing
- GitHub Releases with both APKs
- Firebase App Distribution for beta

Phase 7 | Polish + Play Store

Week 10+

- App icons, splash screens, illustrations
- Error states, empty states, loading skeletons
- Fastlane Google Play publishing
- Store listing: screenshots, description